MDB Calibration Technician Training Module 010

Practical Field Evaluation / Final Scenario Test

Version v0.1 | Internal Training Deliverable

Training Lane: Workspace 006-B Calibration Technician Training

Position: Final module after Module 009 Training Progress, Supervised Runs, and Solo Service Approval

Control item	Current setting
Module owner	Mobile Drill internal calibration training program
Primary evaluator	Senior calibration tech
Formal approval wording	Approved for solo service appointments
Evaluation method	Pass / fail by scenario, with mandatory retraining or retest triggers
External requirement before solo work	PDI onsite training certification completed before any solo service appointments
High-priority eligibility	Separate senior calibration tech judgment after proven field performance over time

This module is designed to confirm practical readiness for real service appointments. It is not intended to keep a capable technician in training indefinitely or judge them against a perfect classroom scenario.

1. Locked Module Purpose

Module 010 proves that the trainee can apply Modules 001-009 together in realistic field situations before being approved for solo service appointments.

This module does not introduce a new technical process. It is the final practical evaluation and readiness checkpoint.

> *Core rule: The trainee must demonstrate safe, repeatable, reviewable calibration work across realistic scenarios.*

2. Real-World Readiness Principle

The purpose of Module 010 is not to create a perfect classroom technician.

The purpose is to confirm that the trainee understands what successful field calibration looks like, can make sound decisions in real jobsite conditions, and can produce usable, reviewable work around people, machinery, and natural ground conditions that will never behave exactly the same twice.

A technician cannot stay in the training stage forever. Once the senior calibration tech is confident the trainee can perform standard service appointments at a high level, the trainee should move into real field production with appropriate monitoring.

3. Source Basis and Training Boundary

- Modules 001-009 of the MDB Calibration Technician Training Lane.
- Workspace 006 Calibration Workspace training-progress and escalation structure.
- Calibration field workflow, analyzer workflow, LP/LE/depth interval rules, data-quality rules, report handoff rules, and customer communication rules established in prior modules.
- SME direction that Module 010 should test readiness for real field interaction rather than classroom perfection.
- SME direction that PDI onsite training certification is the final external requirement before any solo service appointments.

Training boundary: this module evaluates field readiness for standard calibration service appointments. It does not grant high-priority job eligibility by itself.

4. Evaluation Format

Module 010 is graded by scenario. Each scenario is marked:

- PASS
- FAIL
- RETRAIN / RETEST REQUIRED

The senior calibration tech may add notes, corrections, or required retest items.

A trainee does not need to be perfect, but must show safe equipment handling, setup discipline, accurate data documentation, good judgment, proper escalation, professional customer communication, and complete report handoff.

5. Mandatory Retraining / Retest Triggers

The following items do not permanently disqualify a trainee, but they do require correction before approval for solo service appointments:

- Allows the instrumented rod to enter the hollow stem auger or borehole.
- Presses forward after a failed pulse check without correcting the issue.
- Ignores suspected strain gauge damage.
- Does not escalate unresolved failed pulse/ping after backup sensors also fail.
- Fails to record 6-inch blow counts.
- Guesses LE instead of measuring.

- Treats refusal data as normal clean data.
- Promises the customer a pass/fail result or energy percentage before review.
- Sends or promises final report delivery without the qualified report handler.
- Fails to report equipment damage.
- Cannot explain what happened during questionable data.

Training rule: One serious safety or data-quality miss should trigger focused correction. The goal is not punishment; the goal is to make sure the technician understands the consequence and can perform correctly next time.

6. Required Evaluation Categories

Category	Evaluation focus
Job preparation	Understands appointment purpose, rig/customer context, required equipment, and jobsite communication.
Analyzer kit setup	Prepares analyzer hardware, instrumented rod, cables, sensors, power, table, and documentation materials.
Software workflow	Creates/uses the correct data file, enters rig/job/depth context, checks sensors, and ends tests correctly.
LP / LE / depth accuracy	Measures and records LP, LE, depth interval, and sampler travel accurately.
Pulse check troubleshooting	Stops before data collection when pulse check is bad and follows troubleshooting/escalation path.
Data collection discipline	Controls start/stop process, watches first 6 blows, and records blow counts in 6-inch increments.
Refusal / rapid advancement response	Recognizes conditions that affect data quality and documents or stops appropriately.
Equipment protection	Protects instrumented rod, sensors, strain gauges, cables, and connectors.
Documentation and calibration log	Completes one calibration log per rig and captures field notes needed for report building.
Report handoff	Provides complete analyzer data, calibration log, notes, and issue context to the qualified report handler.
Customer communication	Explains calibration clearly, avoids promises, and stays professional under pressure.
Escalation judgment	Knows when to continue, when to rerun, when to document, and when to call for senior guidance.

7. Prerequisites Before Final Approval

Before final approval for solo service appointments, the trainee should have completed the internal training path defined in Module 009:

- Modules 001-009 completed.
- Minimum 5 observed / work-with calibration appointments, or more based on comprehension.
- Comprehension check completed before trainee leads supervised work.
- Minimum 5 supervised service appointments where the trainee does the work while the senior calibration tech observes.
- Minimum 5 observed report builds before historical report run-throughs.
- Minimum 4 historical report run-throughs where trainee results are compared to original reports.
- Trainee-built reports from the trainee's own collected data under senior calibration tech review until performance is acceptable.
- Correction items addressed or actively routed into additional supervised work.

8. Final External Certification Requirement

Before any solo service appointments, the trainee must obtain certification from the onsite training at Pile Dynamics, Inc. / PDI.

The Mobile Drill internal training program is designed to prepare the trainee to enter PDI training with confidence and understand the field meaning behind the analyzer, sensors, data collection, and report workflow.

After completing a meaningful portion of this internal training, the trainee should be prepared to understand and discuss the PDI course content at a high level.

> *PDI certification is the final external step before solo service appointments. Internal readiness and PDI certification must both be satisfied.*

Scenario 1 - Normal Calibration Appointment

Scenario

Customer has one rig ready for calibration. Setup is normal. No refusal, no rapid advancement, and no failed pulse check occur.

Trainee must demonstrate

- Confirms rig, customer, borehole, driller, and tech information.
- Sets up analyzer kit correctly.
- Enters depth interval correctly.
- Measures and enters LP and LE.
- Confirms sensor connection.
- Runs pulse check before live collection.
- Starts collection only after setup is ready.
- Records 6-inch blow counts.
- Ends test properly.
- Completes calibration log.
- Hands off data correctly.

Pass standard

PASS if the trainee completes the workflow cleanly with usable data and complete documentation.

Scenario 2 - Failed Pulse Check

Scenario

The pulse check does not produce the expected graph/ping.

Trainee must demonstrate

- Does not start live data collection.
- Reconnects and checks cables.
- Checks sensor and pigtail connections.
- Restarts software completely.
- Replaces accelerometers/sensors with backups if needed.
- Reruns pulse check.
- Proceeds only if pulse check passes.
- Escalates to office if backup sensors also fail ping/pulse check.

Pass standard

PASS if the trainee troubleshoots in the correct order and escalates if backup sensors also fail.

Mandatory retraining / retest triggers

- Retrain/retest if the trainee starts collection after a failed pulse check.
- Retrain/retest if the trainee keeps guessing after backup sensors also fail.

Scenario 3 - Refusal

Scenario

Sampler does not advance a full 6 inches in 50 blows.

Trainee must demonstrate

- Recognizes refusal.
- Records blow count.

- Records actual advancement.
- Documents depth interval.
- Does not treat refusal data as normal clean calibration data.
- Communicates issue professionally.
- Follows field decision or escalation path.

Pass standard

PASS if the trainee recognizes refusal and protects report quality through documentation and proper handling.

Scenario 4 - Rapid Advancement / Instrumented Rod Risk

Scenario

During the first few blows, the sampler advances quickly and the instrumented rod is at risk of entering the hollow stem auger or borehole.

Trainee must demonstrate

- Watches first 6 blows closely.
- Stops hammer operation before the instrumented rod enters HSA or borehole.
- Protects sensors and instrumented rod.
- Communicates clearly with operator.
- Documents rapid advancement.
- Corrects setup or selects another interval if appropriate.
- Escalates if damage is suspected.

Pass standard

PASS if the trainee protects the instrumented rod and prevents damage.

Mandatory retraining / retest triggers

- Retrain/retest if the trainee allows hammering to continue while the instrumented rod is at risk.

Scenario 5 - Wrong Depth Interval or Missing Blow Counts

Scenario

The trainee entered a 20.0-22.0 ft interval, but the sampler only advanced to 21.5 ft. Or the trainee forgot to record 6-inch blow counts.

Trainee must demonstrate

- Recognizes the interval mismatch.
- Corrects documentation if caught on site.
- Collects another proper data set if needed and possible.
- Records 6-inch blow counts.
- Documents what happened.
- Understands software total is not enough by itself.

Pass standard

PASS if the trainee fixes the issue on site when possible and documents clearly enough for report building.

Mandatory retraining / retest triggers

- Retrain/retest if the trainee says the software total is enough and does not document 6-inch increments.

Scenario 6 - Customer Pressure / Do-Not-Promise

Scenario

Customer says: "We just need the paperwork. Can you tell us if the hammer is above 60 today and get the report sent over?"

Trainee must demonstrate

- Does not call calibration just paperwork.
- Explains calibration as energy-transfer measurement.
- Uses approved plain-language explanation.
- Does not promise pass/fail result.
- Does not promise energy percentage.
- Does not promise same-day report.
- States standard report delivery expectation when appropriate.
- Routes urgency to qualified report handler.
- Protects data quality despite customer pressure.

Pass standard

PASS if the trainee protects Mobile Drill credibility and avoids promises outside authority.

Approved field explanation

"We are checking how much hammer energy is actually getting into the drill rods so your SPT blow counts can be corrected to the N60 standard and used with confidence."

Scenario 7 - Report Handoff Problem

Scenario

Data collection is complete, but the calibration log is incomplete or the report builder cannot tell what happened at one interval.

Trainee must demonstrate

- Identifies missing handoff information.

- Confirms analyzer data file exists.
- Confirms calibration log is complete.
- Verifies depth interval, LP, LE, and blow counts.
- Adds field notes while the details are still fresh.
- Provides photos/video if issue affected calibration data.
- Routes clarification to the qualified report handler.

Pass standard

PASS if the trainee corrects handoff gaps before they become report delays.

Scenario 8 - Equipment Damage Distinction

Scenario

An accelerometer appears damaged during a job. Later scenario variant: strain gauge damage is suspected.

For accelerometer damage, trainee must demonstrate

- Checks for cable or connector damage.
- Replaces accelerometer with backup if available.
- Reruns pulse check.
- Continues only if pulse check passes.
- Calls office and documents damage.

For suspected strain gauge damage, trainee must demonstrate

- Stops.
- Does not treat it as field-replaceable.
- Calls office immediately.
- Routes through drill manager / senior authority path.
- Does not continue without direction.

Pass standard

PASS if the trainee treats accelerometer damage as correctable when backups work, but treats suspected strain gauge damage as immediate escalation.

9. Final Evaluation Decision

After all scenarios, the senior calibration tech marks one outcome:

Decision	Meaning
Continue supervised training	The trainee needs more field time before another final readiness review.
Retrain / retest required	Specific module, scenario, or field skill must be repeated before approval.
Approved for solo service appointments	Trainee may perform standard solo calibration service appointments after internal approval and required PDI onsite certification are complete.
High-priority eligibility	Not automatic. This is a separate senior calibration tech decision after proven performance over time.

The senior calibration tech may require retest, additional supervised appointments, or additional review when needed.

Approval is based on demonstrated readiness for real standard service appointments, not perfect performance in every imagined scenario.

10. Final Evaluation Record

Record field	Entry
Trainee name	
Senior calibration tech	
Evaluation date	
Evaluation location / job reference	
Modules 001-009 completed	Pass / Needs review
PDI onsite training certification completed	Yes / No / Pending
Scenario 1 - Normal appointment	Pass / Fail / Retest
Scenario 2 - Failed pulse check	Pass / Fail / Retest
Scenario 3 - Refusal	Pass / Fail / Retest

Record field	Entry
Scenario 4 - Rapid advancement	Pass / Fail / Retest
Scenario 5 - Depth/blow-count issue	Pass / Fail / Retest
Scenario 6 - Customer pressure	Pass / Fail / Retest
Scenario 7 - Report handoff problem	Pass / Fail / Retest
Scenario 8 - Equipment damage distinction	Pass / Fail / Retest
Mandatory retraining / retest trigger noted	Yes / No
Correction items required	
Final decision	
Approved for solo service appointments	Yes / No / Pending PDI certification
High-priority eligible	No / Later review / Approved separately
Senior calibration tech final approval note	

11. Knowledge Check

#	Question	Answer key
1	Is Module 010 mainly new teaching content or evaluation?	Evaluation.
2	How is Module 010 graded?	Pass/fail by scenario.
3	What happens if backups fail the pulse check too?	Likely cord/connector issue; escalate to office.
4	What is refusal?	Sampler does not advance a full 6 inches in 50 blows.
5	What should happen if instrumented rod risks entering HSA/borehole?	Stop hammer operation and protect the instrumented rod.
6	Is software total enough without 6-inch blow counts?	No.
7	Can trainee promise the hammer is above 60 before review?	No.
8	Who handles final report delivery?	Qualified report handler.
9	Is high-priority eligibility automatic after solo approval?	No. It is a separate senior calibration tech decision.
10	What is the final approval wording?	Approved for solo service appointments.
11	What external certification is required before solo service appointments?	PDI onsite training certification.

12. Practical Evaluation Checklist

Signoff item	Pass standard	Result
Normal workflow	Trainee completes standard appointment cleanly.	Pass / Retest
Failed pulse check	Trainee follows troubleshooting path and escalates correctly.	Pass / Retest
Refusal	Trainee identifies and documents refusal correctly.	Pass / Retest
Rapid advancement	Trainee protects instrumented rod and stops hammer if needed.	Pass / Retest
Depth interval accuracy	Trainee corrects interval/documentation issues.	Pass / Retest
Blow-count documentation	Trainee records all 6-inch increments.	Pass / Retest
Customer pressure	Trainee avoids promises and protects data quality.	Pass / Retest
Report handoff	Trainee provides complete files/log/notes.	Pass / Retest
Equipment damage	Trainee distinguishes accelerometer vs strain gauge response.	Pass / Retest
Final judgment	Trainee shows safe, repeatable, reviewable work.	Pass / Retest
PDI certification status	PDI onsite training certification is complete or pending before any solo appointments.	Complete / Pending

13. Revision History

Version	Change	Notes
v0.1	Initial controlled deliverable for Module 010.	Created after SME approval of pass/fail scenario format, softened retraining language, real-world readiness principle, no signature requirement, and PDI onsite certification requirement.

14. Completion Note

Completion of Module 010 rounds out the 001-010 internal calibration technician training module set. The next control artifact should be a Calibration Training Module Register / Master Index that lists every module, version, status, purpose, and approval/control notes.